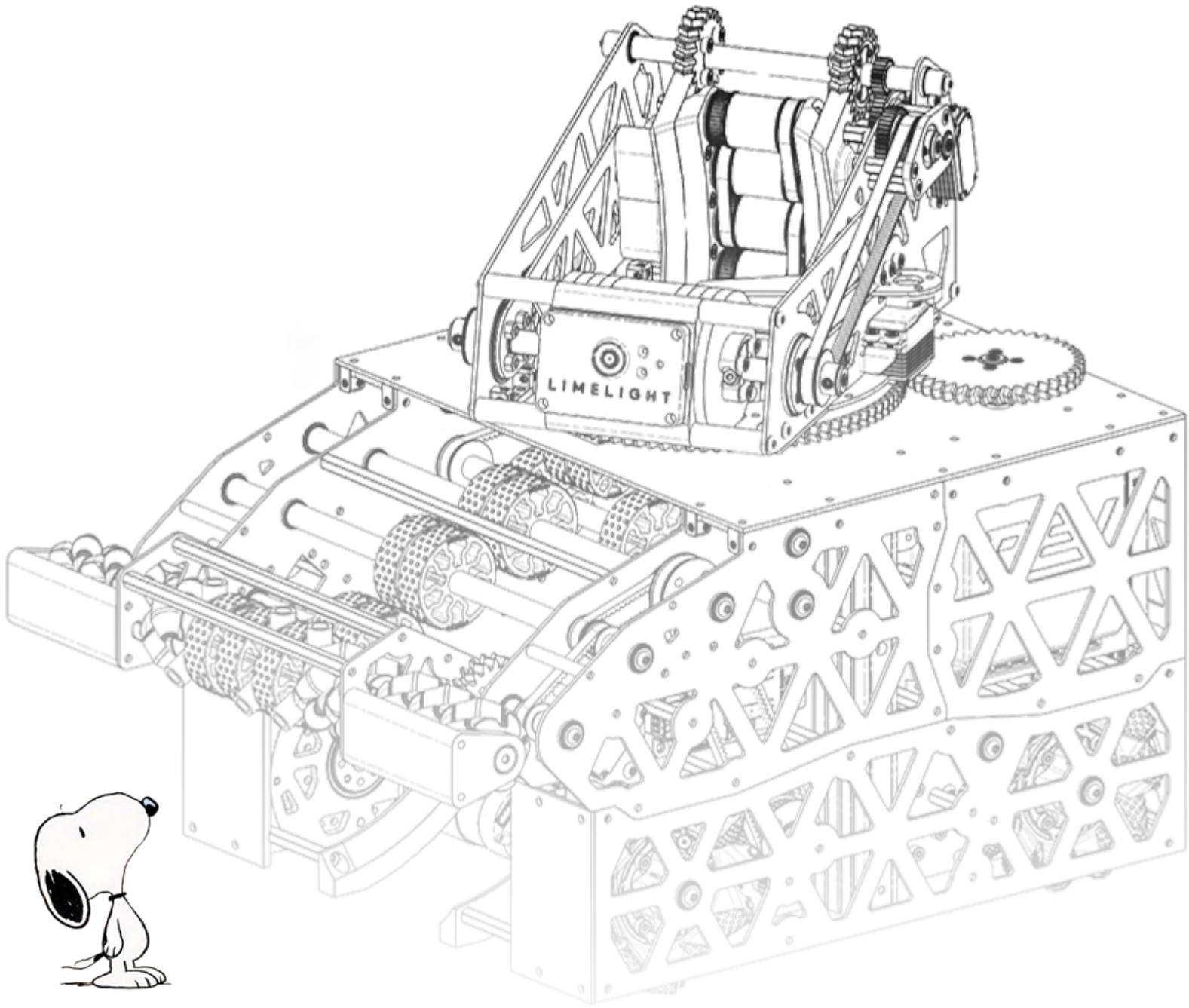




Verge Robotics Presents...

SNOOPY

Engineering Portfolio



Team Verge #23250

With **clear goals** set for each team member, new and old, and several industry **professionals** that helped with the growth of ideas in our team, our team has become an **active member** of the **STEM community** around us.

Sustainability Strategy

Team Expenses

- **Diversified** revenue to avoid dependence on one single source of finances by obtaining ...
 - **team fees (\$2600)**,
 - **grants (\$2200)**,
 - **door-to-door fundraising (\$820)**,
 - **local business sponsorships (\$1300)**
- **Saved ~\$1150** from reusing parts from previous seasons and using a wooden chassis
- Used a **Gantt-style timeline** to keep track of season design goals

Team Roles & Leadership Development

- Every member has a role with personal goals → Look @ page 2 of the Engineering Portfolio
- **Recruited 5 rookies** to succeed veteran members
- Rookies were given real leadership opportunities
 - Harsha, Keerthi, & Anvika **mentored the FLL team Golden Falcons to a 2nd place Champion's Award** and encouraged to do door-to-door fundraising to build public speaking skills

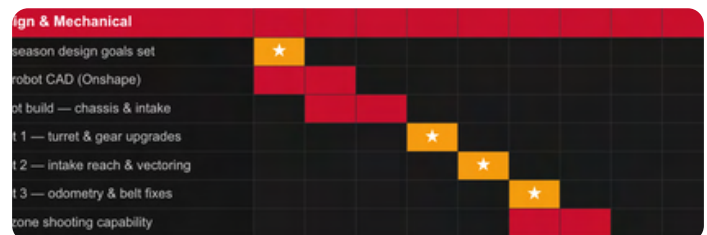
Management of Risks/Constraints

- Used a **Gantt-style season roadmap** alongside **decision matrices** to weigh trade-offs between fixes
 - Ensured that we were reasonable in what we spent
- Ensured **long-term team sustainability** through continuous recruitment
 - Maintained an **Edge FLL → Verge FTC pipeline** so as veterans graduate, incoming rookies enter with foundational knowledge

Google Sheets Expenses Tracking

Team Budget Summary			
Category	Amount	Category	Amount
Revenue		Expenses	
Team Fees	\$2,600	Game Set & Registration	\$1,2
Door-to-Door	\$820	Robot Parts	\$8
Grants	\$2,200	Outreach Materials	\$7
Fundraising/Local Business	\$1,300	Food	\$1
Total Revenue	\$6,920	Total Expenses	\$2,8
Remaining		\$4,100	

Gantt Style Timeline



Sponsorship Strategy

- **Reached out on LinkedIn directly** from team members' profiles to CSR (corporate social responsibility) managers
- Sent **personalized impact reports** after each season to show sponsors where exactly their money went (e.g. # of kids reached/# of awards), keeping them a sustained sponsor
- Used door-to-door fundraising from St.Albans to Ladue to **sustain local business partnerships** (e.g. Plaza Orthodontics & Bombay Bazar)

Sponsors



Member Goals

Critical Thinking Onshape Java **4th Year****Vivek Maddineni - Captain**

Don't overcomplicate design goals, and make use of master sketches in Onshape.

Android Studio Patient Java **4th Year****Rohan Gopalan - Prog. Lead**

Keep software simple and make better use of Git to keep track of progress.

Design Onshape Creative **4th Year****Ahan Singh - Vice Captain**

Learn how to make durable 3D-printed parts that can last a season.

Optimistic Organized Good Speaker **4th Year****Abhi Boga - Business / Outreach**

Find new ways to connect with sponsorships leading to further awareness of FIRST.

Resilient Controlled Tactical **4th Year****Arvind Sunilkumar - Driver / Business**

Manage finances cleanly and make ample time for driver practice.

Collaborative Patient Onshape **4th Year****Pardhu Konakandla - Build / Outreach**

Learn how to talk in an impromptu manner at outreach and sponsorship presentations

Confident Onshape 3D-Printing **3rd Year****Adit Swami - Build / Programming**

Gain experience packaging hubs, motors, and servos effectively in a robot.

Critical Thinking Onshape Java **Rookie****Sai Kondamadugula - Build / Programming**

Research game strategy and penalties to inform design and code considerations.

Problem Solving Java Android Studio **Rookie****Harsha Konakandla - Programming**

Learn how to incorporate math and physics principles into code to enhance performance.

Critical Thinking Resourceful Positive **Rookie****Anvika Malugu - Outreach**

Brainstorm and organize several fundraising events around the community.

Resourceful Committed Proactive **Rookie****Keerthi Maddineni - Outreach**

Take initiative in mentoring rookie FLL teams to success throughout the season.

Creative Collaborative Persuasive **Rookie****Sachit Prakash - Outreach**

Design a sponsorship outreach kit including email templates and impact report.

Team Goals

- **Recruit** and **integrate** 5 rookie members well into the team and the FIRST community
- Improve **consistency in practice time** before competitions
- Gain **new sponsorships** to support robot development and outreach goals.
- Qualify for the **FIRST World championship** with the **Inspire Award** or **robot performance**.
- Develop strategies to **adapt quickly during competitions** or unexpected challenges

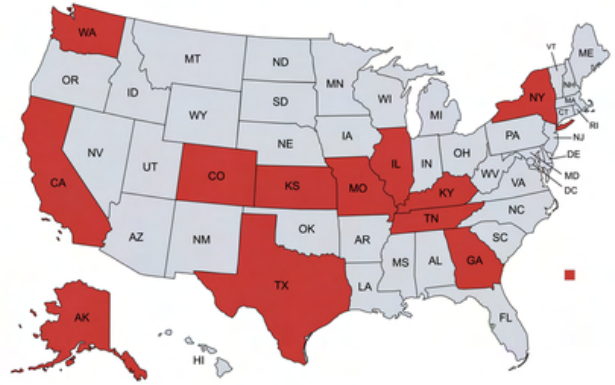
Motivating and Mentoring FTC

Verge

FTC in America

At Robot Rumble, an off-season event, we met many **new Illinois teams**. In preparation for this event, we contacted teams **across the country** for advice in **computer vision** and **design considerations**.

#11260 Up-A-Creek
#18139 Rebel Robotics
#21336 I Forgot
#4348 RoboKnights
#23521 DeSoto Technix
#25679 The Cosmobots
#27971 Null Pointer
#4780 Robolution



FTC across the World

We sought advice and **actively mentored** rookie & veteran teams **around the world** via Instagram/Discord. For team #33680, we were their **first international connection**.

#22589 Marvels of Nigeria
#12993 RoboKings Aurum
#15972 TehnoZ
#19666 Forever Knight
#33680 Atomic Heart
#18763 Texpend
#23504 Space Tech
#23435 Gyrobotic Droids



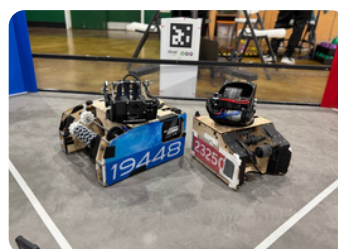
GOLDEN FALCONS MENTORSHIP

Mentoring the Next Generation

- New members **Harsha, Keerthi, & Anvika** mentored an FLL Rookie team, **Golden Falcons**, who then won the **2nd place Champion's Award & Robot Performance**, qualifying for the Missouri Regional Championship
- We are proud of our rookies' **leadership** and are excited to see them **continue to inspire**.

The Verge of Winning

- We mentored/scrimmaged with **X Bots**, **Gluons**, **Baryons Luxons**, **Action Robotix**, and **AutoBotz** several times throughout the season.
- We taught **Luxons** about bare motors and flywheels. We taught **Action Robotix** and **AutoBotz** how to use **Pedro Pathing**.
 - Under our guidance, **Luxons and Action Robotix** both **qualified to states**. Action Robotix was even **top 30 in MO/KS** at one point. Autobotz, a rookie team, **qualified to superqualifiers**.

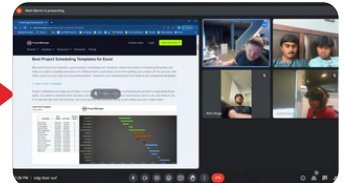


Team Plan – Mentors

Verge



- Matt Martin (WashU MBA Grad) → Gained through a family connection
 - Introduced us to **Gantt-Style Timelines** to improve tasks tracking & **flowcharts** to accelerate ideation



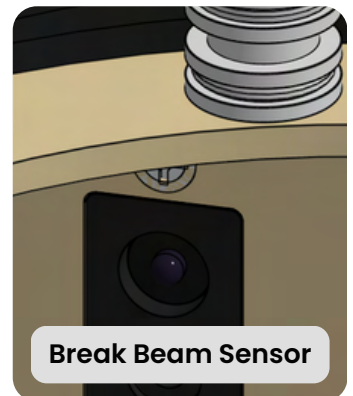
- Dr. Sarangapani: S&T Prof. & IEEE Member → Gained via cold email
 - Introduced us to **ferrite chokes** to reduce pinpoint, controller disconnects, & helped improve our loop times by simplifying code



- Ajay Sunilkumar: FRC/Purdue Eng. Alum → Gained via a family connect
 - Helped streamline development timeline & showed us the power of **variable driven CAD**



- Dr. Schuh: UIUC ECE Eng. Prof. → Gained via a FIRST Alumni at UIUC
 - Taught us to use **velocity for the D term in PIDF**, increasing stability
 - Suggested to use **break beam sensor** in V2



- Maxwell Geiger: S&T PhD Student → Gained by Dr.Sarangapani at S&T
 - Helped **optimize Scarlet curve** following with his expertise in self driving cars

FIRST Knowledge in Action

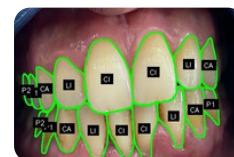
SmileScan

Problem:

- Dental plaque detection mostly relies on **uncomfortable disclosing agents**
- AI solutions like **YOLOv7** reach **89% accuracy** but still need disclosing agents
- Current solutions are too **costly or inaccessible**

Solution:

- SmileScan— a software which scans teeth without agents, with an error rate lower than dentists.
- Built with **Python, OpenCV, PyTorch, Firebase**
- Delivers accurate plaque metrics to **underserved populations** without costly equipment or discomfort and is being tested at **2 dental practices** in rural India.



STEP 1



STEP 2



STEP 3



- During a presentation on STEM careers to **a class of 40 girls in rural India**, one student raised her village's struggle with **expensive** dental care and **lack** of awareness surrounding **dental health**. Inspired by the student's insight, **SmileScan was born**. Through collaboration with two dental clinics in a nearby town, we obtained **552 sample images** to train our AI model.

St. Louis Mayor's Office Timeline

- Met with the **St. Louis mayor's office** to discuss **outreach strategies** and an **outlet** to reach more people, effectively increasing **STEM awareness**
 - Conducted **6 bi-weekly sessions** with elementary/middle schools in the **Florissant & Ferguson School District**, an disadvantaged school district, introducing **FIRST principles**. We served at the **Saturday Success Program at Way Center** for 2 months—program for underserved children.
 - Helped **5 kids** even join FLL teams in the STL region
 - Created a platform, **Numeria**, to improve math proficiency



Griffith
Elementary



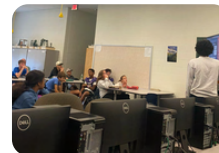
Way
Center



In an effort to increase math proficiency in the Florissant school region, **currently 16%**, we developed **Numeria** — a simplified math learning platform designed to improve arithmetic skills

Elementary Schools

- We went to **12** different elementary and middle schools, **inspiring ~450 students** about **FIRST and STEAM principles** (incl. 2 Indian schools)
- At both Blevins, we **sparked interest** in nearly **25 kids** in starting ~ **2 FLL teams** sponsored by the Rockwood School District



Reach Beyond Ordinary

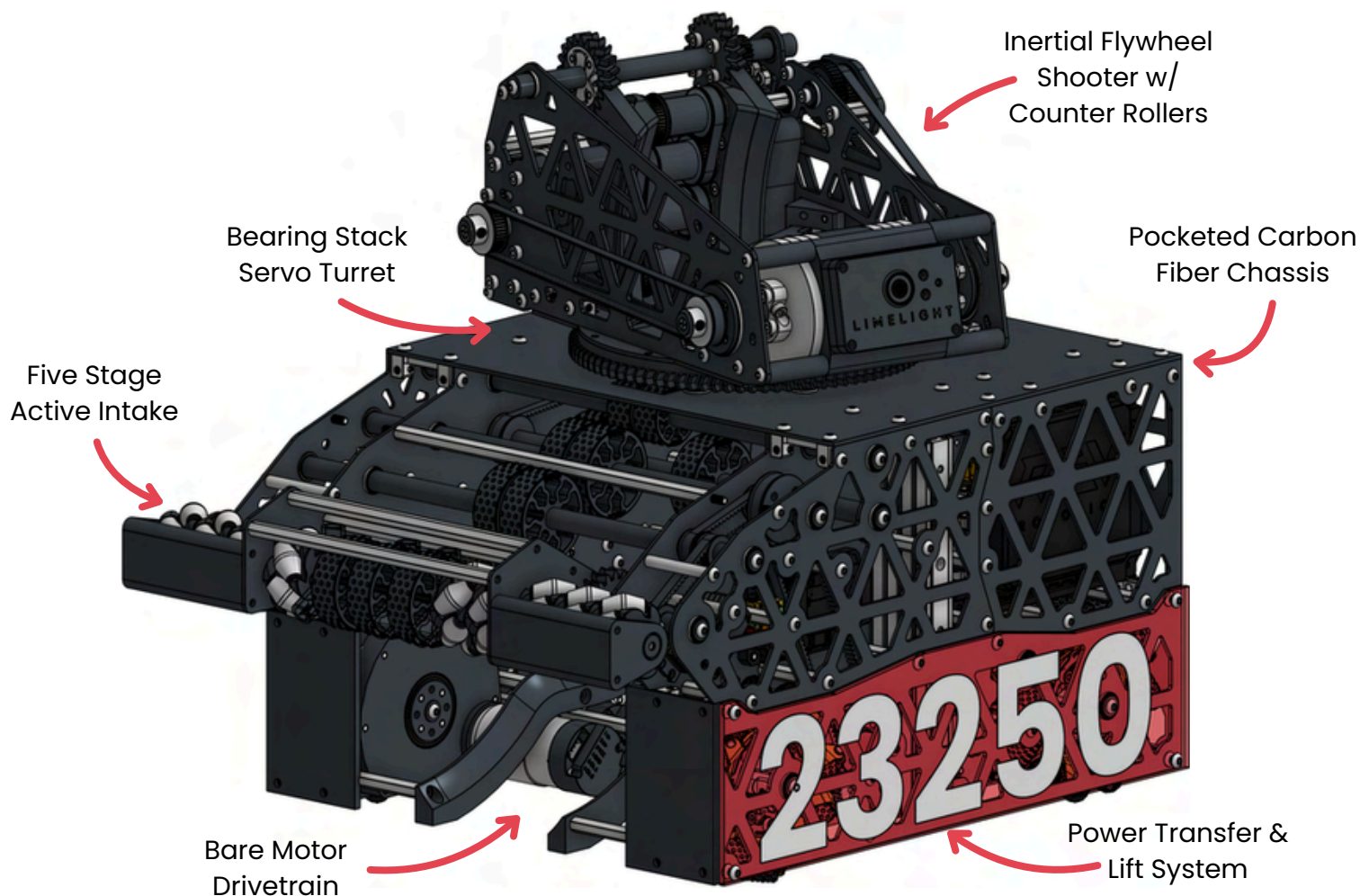
- Sent over **a dozen goodie bags** with 3D printed toys and STEM gifts to foster children over Christmas break
- Held **bi-weekly sessions** for a month where Vivek helped a child with autism build his first own PC.
- Over the season, Vivek, Pardhu, and Hritika have **visited 3 special educational schools**, teaching these kids about the basic **engineering design process** as improving **problem solving skills**



Print to Partner

By connecting with our filament sponsor, PolyMaker, we were able to send **15 packages** of filaments to teams **#22589**, **#33680 (Rookie team)**, and **#19666** in **Nigeria, China, & Kazakhstan**

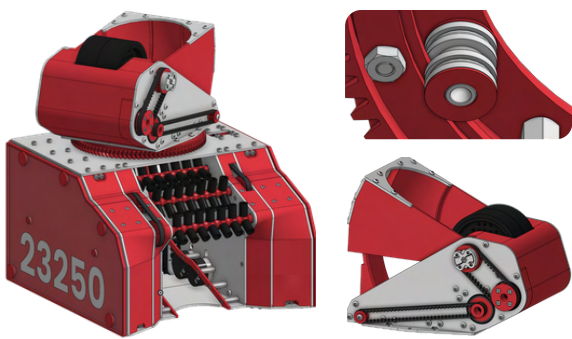




Our Design Philosophy

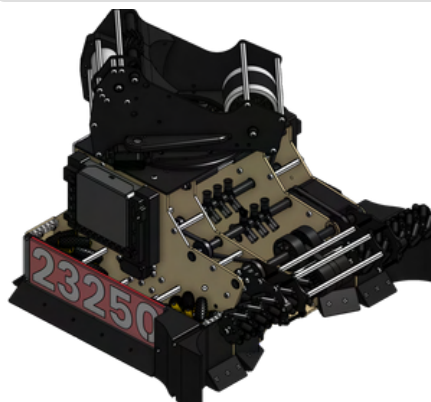
Instead of building the robot around an indexer or double park system, we **fit parking where we could** and rather prioritized **sheer functionality and reliability**, primarily optimizing **intaking, shooting, and defending**.

Snoopy V1.5



Snoopy V1 **reliably** hit a **12-ball** autonomous by meet 2 and ranked **1st in MO/KS** by qualification rankings after qualifiers. A complete rebuild was needed to improve performance at later competitions.

Snoopy V2.5



Snoopy V2.5 **enabled** us to be the **Winning Alliance** Partner and **Control Award** winner at the Missouri/Kansas championship. Its major changes included higher intake **efficiency**, a weighted flywheel, and **defensive** wedges.

All custom components are designed in **Onshape**, with **Blender** handling final renders and visualizations.

Pre-season

Design Goals

- Maintain a **small** footprint
- **Avoid complicated** systems/code
- Focus on “touch it, own it” intake
- Ensure **consistent** close zone **shooting**
- Ignore **double parking** or **indexing**

We **nailed** our design goals, had no large changes, and focused on **optimizing**

Progress Tracking

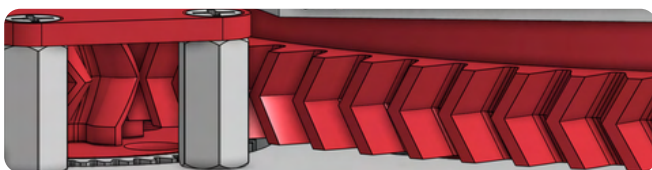
We **tracked** our **progress** between each league **meet** by counting the **number** of artifacts scored each match and taking an **average** of that number from every **match** at the meet. We compared this number to the **averages** from **past** league meets.

Meet 1

V1.3

Changes Made

- Swapped MOD 1.0 turret gears for **3D printed MOD 2.0 gears** for reduced skipping and improved motor torque transmission.
- Changed vectoring guide geometry to **reduce jamming** and **increase vectoring efficiency**.
- Increased tolerance inside bearing stack of turret to **decrease friction** while retaining **maximum rigidity**.



Analysis

Worked Well:



- New MOD 2.0 gears were extremely reliable
- Adjusted bearing stack tolerances decreased load on motor.

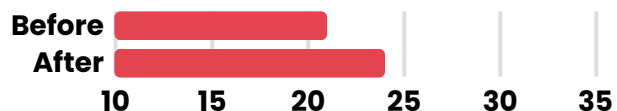
Needs Improvement:



- Unreliable transfer; balls get stuck in the deadzone.
- Color sensor did not trigger reliably

Prototype vs Meet 1

Artifacts/Match



Worked Well:



- Surgical tubing made intake much more consistent.

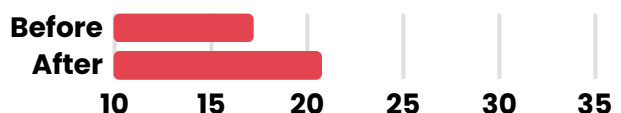
Needs Improvement:



- Odometry setup was inconsistent and cost us two matches.

Meet 1 vs Meet 2

Artifacts/Match



Meet 2

V1.4

- Added **longer surgical tubing** and goBilda GripForce gecko wheel rollers on end of intake for **farther reach** and **faster vectoring**.
- Further **optimized ramp geometry** to ensure **smooth transfer action** made possible by this bump.



Meet 3

V1.4

Changes Made

- Added a thin layer of **aluminum** tape to the shooter to **reduce backspin** and **mitigate** bounce outs.
- Attached a ferrite choke to our **odometry** setup's i2c cable.
- Gobilda 4-bar odometry pods are usually **very accurate** but prior to this meet we added some tape near the connector to **increase reliability**



Analysis

Worked Well:



- Aluminum tape slightly reduced backspin
- Ferrite choke and tape stopped our odometry drift issue.

Needs Improvement:



- We were only able to shoot from one point and many teams played defense against us because of this fatal flaw.

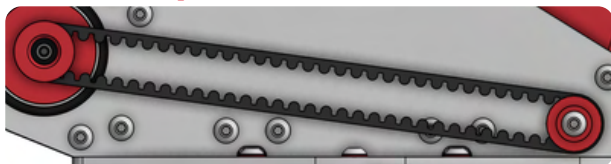
Meet 2 vs Meet 3 ■ Artifacts/Match



Leagues

V1.5

- Screws were **loose** all over the robot so we **reassembled** large parts of it to mitigate any **future** problems.
- HTD5 Belts with tiny gears caused lots of **efficiency loss**, so we **reduced** the pulley side by one tooth to offer less tension and more **efficiency**.

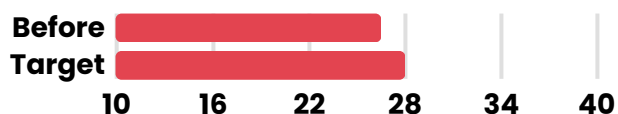


Worked Well:



- 12 artifact autonomous** with approximately **11/12** or **92%** average accuracy on red and blue side.
- TeleOp macro **automatically** drives to point and shoots.
- Far zone working **consistently**

Meet 3 vs Leagues ■ Artifacts/Match



Lessons Learned

- The **first** version of our robot was extremely **effective** early season, but it quickly became **obsolete** at the STL-MID league tournament.
- It's **small** intake, **hard** to maintain design, **constant** tipping, and **unreliable** wiring were the **biggest** pain points that we **resolved** in V2
- We **fixed** this in V2 by **incorporating** an extremely **wide** vector wheel intake
- The **tipping** problem was completely **fixed** when we moved all **6** bare motors to the **bottom** of the chassis, made possible by **crown gears**.

Superquals

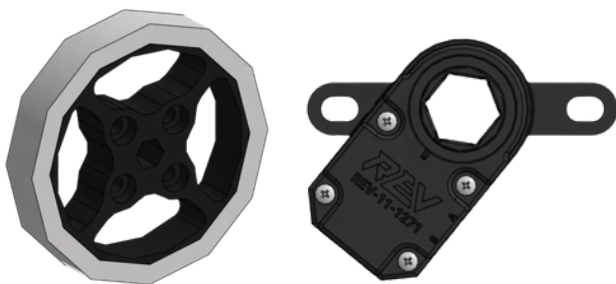
V 2.4

Changes Made

- Extended the intake for **farther** reach and more reliable acquisition.
- Added side **vector wheels** to center game elements and **improve intake efficiency**.
- The new intake is roughly **1.3× faster** than the **previous** design, **reducing** cycle time.



- Reduced** flywheel diameter and added lead tape to **increase** rotational **inertia** and **lower** **compression**.
- Removed grip tape** from vector wheels, resulting in **quicker** intaking.



- Improved** the mounting setup for the **REV through bore encoder** to **minimize play** in measuring the angle of the turret.
- Increased flywheel mass** from 6 loops of lead tape to 18 loops of lead tape in total across **two rings**.
- Added **hot glue** to ensure **consistent connections** in comp.

Analysis

Works Well:



- Lead tape** increased flywheel inertia and **outtake** sizing allowed for faster cycle times
- Increased** intake length allowed us to intake from range **minimized mistakes**

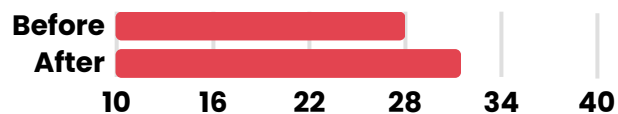
Needs Improvement:



- There are **deadzones** in the intake, interfering with the full intake-outtake sequence
- Shooter **RPM** is too high and drains the battery quickly

Leagues vs Superquals

Artifacts/Match



Works Well:



- Lower** flywheel diameter and compression gave a **more optimal arc**
- Faster intake speed** was **very helpful** in the MO/KS **finals match**.
- Improved** intake geometry **reduced** jamming to be mitigated by drivers.

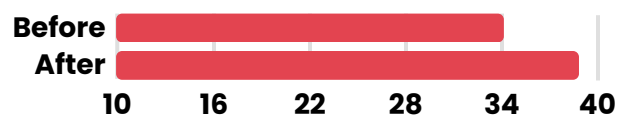
Needs Improvement:



- Intake still has very **small dead zones** despite significant iteration. The ball **no longer gets stuck**, but it can **slow down** at these points in the intake.
- Shooting on the move is **limited**.

Superquals vs MO/KS

Artifacts/Match



MO/KS Championship

V 2.5

V 3.2 World Championship In Houston

Our robot has **improved** by **leaps and bounds** since our first league meet in November, and for the **world championship**, we took it to the **next level** with our **third iteration**. While we kept the aspects of our robot that we **painstakingly iterated** to perfection, we also strived to **improve** the weaknesses. See our **newest additions** to Snoopy below!

Pocketed Carbon Fiber



Until now, we've used **1/8th inch Plywood** for rapid prototyping. For worlds, we've switched over to **2mm pocketed carbon fiber** for the perfect mix of **rigidity and strength**.

Molded Vector Wheels



These **custom-molded** vector wheels **increase grip by 3x** (static friction coefficient) compared to our old **PLA roller** based wheel design.

HTD3 & GT2 Belts



In favor of **efficiency**, we've switched over to **HTD3** and **GT2** belts for our **high-speed intake**, transfer, and shooter subsystems. This brings our effective RPM closer to the **theoretical max output**.

Active Counter-Rollers



To reduce our **bounce-out issues** we opted **reduce backspin** with **actively powered** counter-rollers sped up to **twice the flywheel speed**. Balls now land **cleanly** in the goal.

Bare Motor Pinion Gear



Our new **1/8th inch bore** pinion gears fit much nicer on the 5000 series bare motor's output shaft than our original technique which involved an **adapter piece**. We now have smooth meshing and less noise.

Development Design Process

Brainstorming and Design 1

- Came up with **rigid** build criteria for how we would want our robot to be built and **function**
- Created a model of **full** robot in CAD prior to starting to build to ensure **smooth** building

Building and Testing 2

- Used **custom** and 3D-printed parts to **maximize** customization and **ideal** function
- Practiced** with other **teams** to help visualize how the robot would function on the **field**

Reflection and Redesign 3

- After** competitions, look at and **analyze** parts of our robot which need **improvement** and come up with ways to improve
- Remake** the part on CAD before rebuilding to **confirm** integration with the robot

Chassis Overview

Key Features



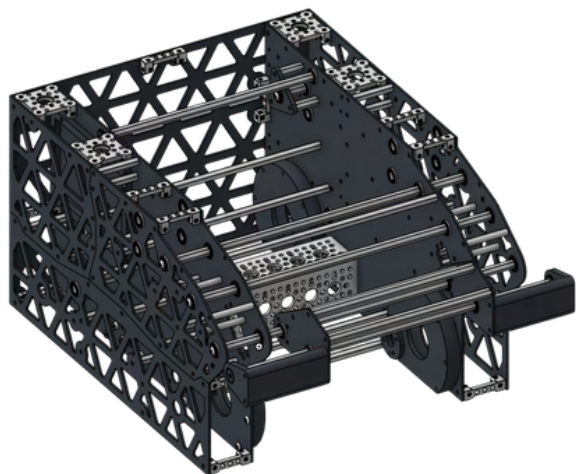
Separated side plates & several **access holes** allow for easy maintenance



2mm pocketed **carbon fiber** reduces weight while maintaining **rigidity**

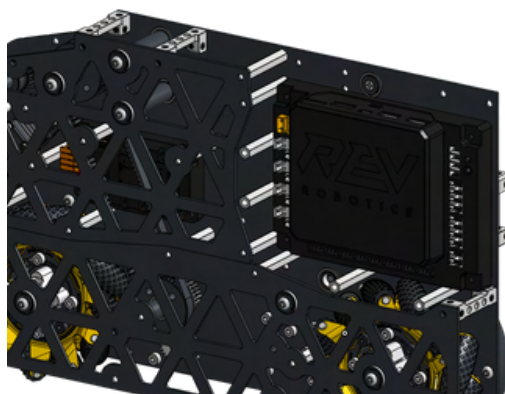


Ratcheting power transfer from drivetrain to lift for **double BASE**



Maintenance

The side plates are split into **three panels** to minimize **unnecessary disassembly** time while preserving structural **strength and reliability** during matches.



Power Transfer System

The lift system **shifts torque** from the rear drivetrain motors to the linear slides. It uses a **clutch** that pushes a sonic hub into a pulley by **using a spring** in the middle. This allows us to use **high torque** motors for the lift instead of adding another motor.



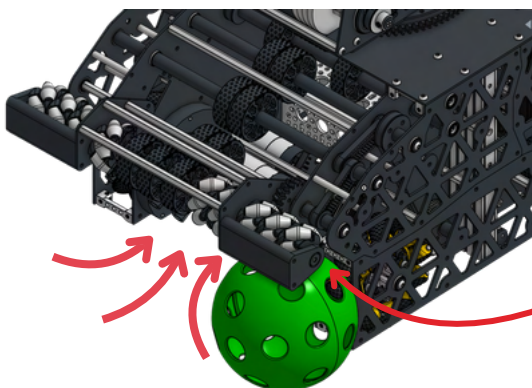
Intake Overview



2600 rpm for first stage and **1500 rpm** for other stages allows for **"touch it, own it"** intaking



Rotating intake arm gives **adjustable compression** and **improved** center of mass



Vector wheels apply a **sideways rolling force** to the artifact, **guiding** it to the center.

Bare Motor Chassis

- **All** the **motors** on our bot have their **gearbox** removed and feature **fully custom** reductions.
- As a result, we **save ~3.3 pounds** throughout the bot



Turret Overview



Innovative **servo input value per radian calculation** gives perfect turret **aiming**



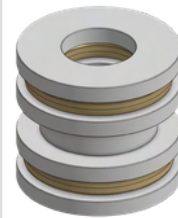
Herringbone gears and **bearing stack** gives **ample torque** for rotation



$$(\text{gear ratio})^{-1} \cdot \frac{\text{servo position range}}{\text{servo angle range}}$$

$$\left(\frac{48}{20} \cdot \frac{41}{89}\right)^{-1} \cdot \frac{1}{355^\circ \cdot \left(\frac{\pi}{180}\right)} = 0.1459785$$

Gear ratio calculation allows code to **accurately** track turret in radians by using servo input

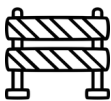


Bearing stack is made of two **thrust bearings** that sandwich an **axial bearing**.

Shooter Overview



Velocity-matched counter rollers cancel backspin of balls, **reducing bounce-outs**



Stopper accounts for **direction of forces** to reliably stop balls from getting shot early



Increased **rotational inertia** of the flywheel to reduce **regeneration** time when shooting



- When balls push against the stopper, it is **off-axis** from the servo's rotation, so the stopper can **rigidly stop balls**.
- Additionally, the closed position is **offset** toward a 3D-printed wall which further **supports the stopper** when it is closed.

Effectively, the hood is a **rack-and-pinion** on a servo, meaning it requires **very little current** to control the **trajectory** of balls all across the field.



$$I = \sum m_i r_i^2$$

$$\tau = I\alpha$$

Balls apply a **torque** and the increased **inertia** leads to less **deceleration** in the flywheel



Counter rollers match **tangential velocity** to reverse the backspin torque on launched balls, making them **far more stable** in the goal.

$$v_T = \omega r$$

Using 8x SWYFT masses, we were able to increase inertia by **8x480 g·cm²**, allowing for consistent **rapid fire far zone** shooting.



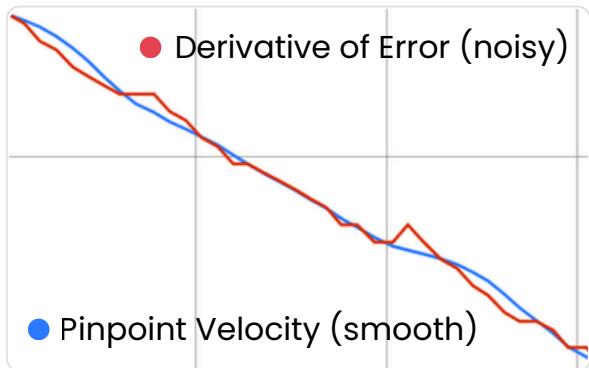
Scarlet – Our Custom Pathing System

Verge

Why Custom?

- Initially when we used **Pedro Pathing**, we were **maxed out on time** for our **12 ball auto**. We knew there were ways to improve, but Pedro limited us heavily.
- Scarlet allows us to **control and observe** each calculation, **optimizing** any place where improvements can be made.

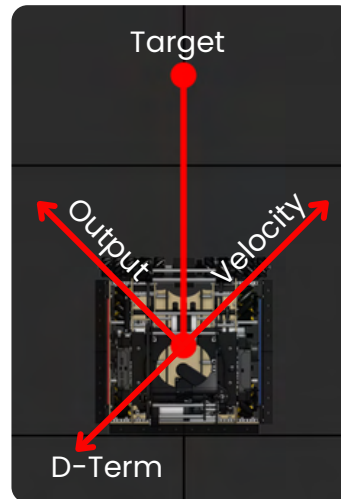
```
antiTip = antiTipBraker.getScalar(errorVec, velocityVec)
drivetrain.setPowers(drive * antiTip, strafe, turn)
```



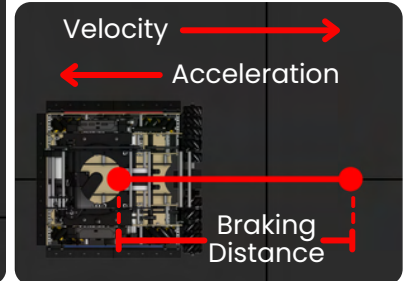
```
autoShoot = new SequentialCommand(
    new InstantCommand( () -> scarlet.resume() ),
    snoopy.shootNear( new Pose(42, 104) ),
    new InstantCommand( () -> scarlet.start() )
```

Impacts & Performance

- After tuning our existing 12 ball auto for Scarlet, we ended up with an **extra 9 seconds**. We used this time to make **synced autos** with teams in our region.
- We made a **teleop macro** to **automatically move** to the shooting position and shoot, which helps against defense and **maximizes cycle times**.
- We also have **shared Scarlet with other teams** and plan to share it on the FTC Discord after more **testing throughout the off-season**.

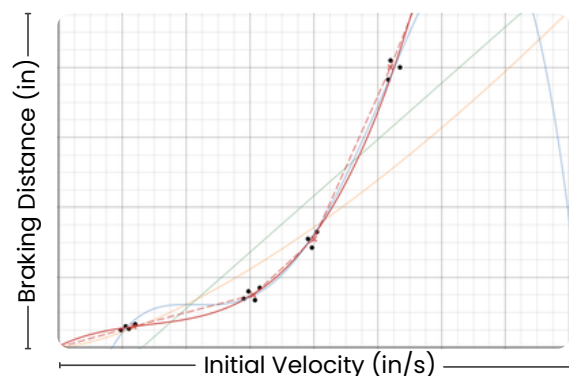
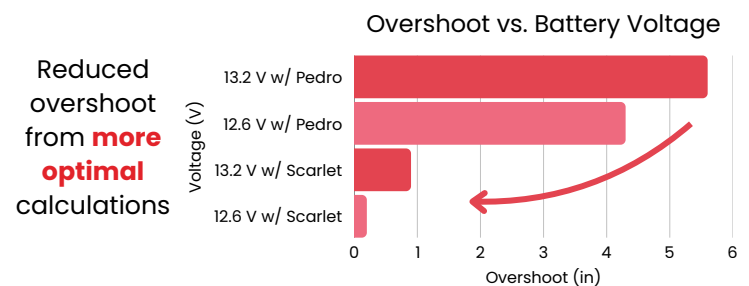


$$P_x = e_x k_P \cos \theta$$
$$D_y = -v_y^2 k_D \sin \theta$$
$$O_x = P_x + D_x$$



Innovative Calculations

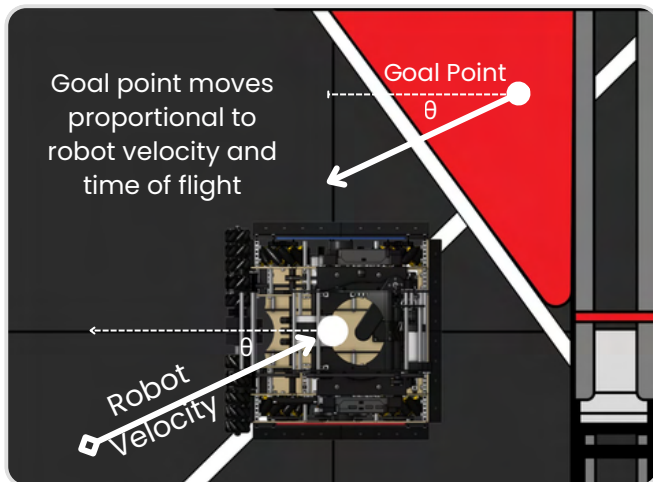
- We noticed that the PID Derivative calculation was **very noisy**, so instead we used the pinpoint velocity which was **far more stable**.
- Also, our V1 robot had lots of **tipping issues**, which meant we had to slow down paths. As a more **optimal solution**, we tuned an **anti-tipping scalar** proportional to the velocity.
- When we tested **predictive braking** from Pedro Pathing, it worked well but we wanted a **regression system** that could more **closely fit** the robot based on the **tuning data**.



Tuning tests several **regression types** to find the **best fit** for Braking Distance vs. Initial Velocity

ActiveArc

- Early on, we could only shoot in one spot, leading to **easy defense** on us. So, we made a system to **shoot anywhere on the field**.
- We tuned **hood and velocity values** at each distance and determined that **linear** was the best fit, so we implemented the **regression equation** in the code.
- This worked so well that even when we extrapolated to shooting **far zone**, we only had to **adjust slightly** to make it work **consistently**

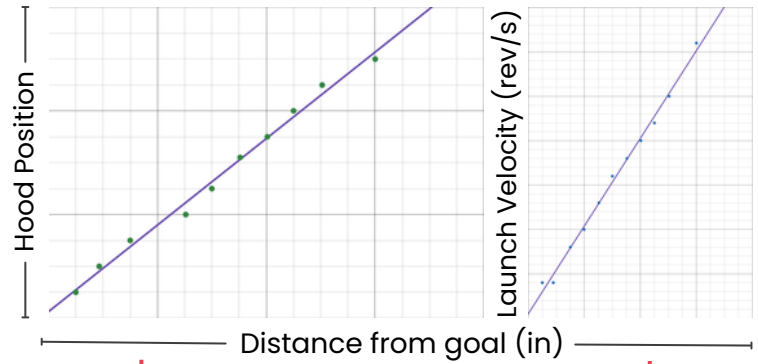


$$\Delta x = v_o t + \frac{1}{2} a t^2$$

```
time = lookupTable.get(getDistanceFromGoal());
offset = velocity * time + 0.5 * acceleration * time * time;
goalPoint = actualGoal - offset.getPoint();
```

Velocity PIDF + Bang Bang

- Many teams use **constant power** set to their flywheel motors, but this can be **inconsistent** with varying battery voltages. So, we opted to use a **tuned PIDF loop** to hold the **velocity stable**.
- In order to **minimize the regeneration time** of our flywheel during shooting, we combine a **Bang Bang controller** with PIDF. This makes the flywheel output max power when its error velocity is **under 100rpm**.



```
double distance = getDistanceFromGoal().magnitude;
double velocity = (4.93968 * distance) + 837.58439;
double hood = (0.00834749 * distance) + 0.0790207;
```

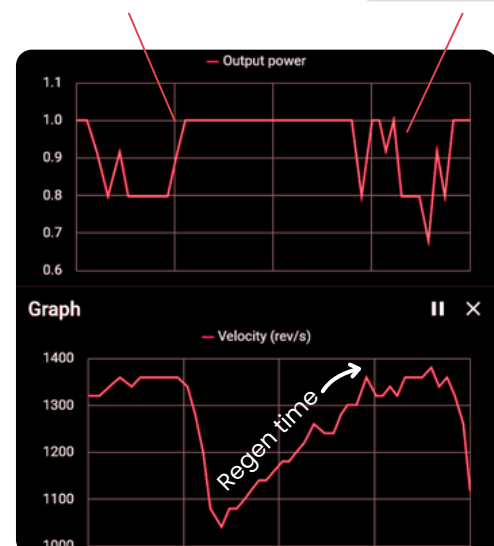
Velocity increases by 5 ticks per **second** and the hood goes up by **0.8%** for every **inch** from the **goal**

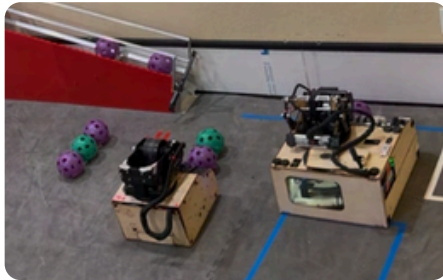
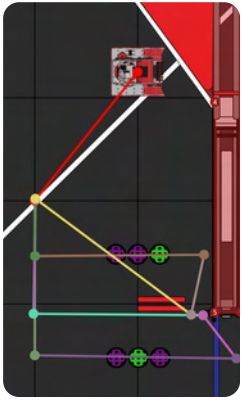
Run 'n Gun

- A major **detriment** to our cycling is the fact that we have to **slow to a stop** when shooting.
- By accounting for the robot's **velocity and acceleration** and the artifact's **time of flight**, we could dynamically move the goal point in code to **predict** where the ball will shoot, and **correct it with the turret**.
- As a result, teams have a harder time playing **defense** and cycling is **far smoother**.

Spin-up time **minimized** with Bang Bang control

Returns to **PIDF** to hold flywheel velocity **steady**





Path visualizer allowed us to quickly create synced autos.

Auto

```
snoopy.shootNear(shootOptions),
intakeAndShoot(84, 14),
snoopy.scarlet.driveTo(PARK_POSE)
```

Teleop

```
shoot = snoopy.shootNear();
controllers.bind(BOTH_DRIVERS, CROSS, shoot);
```

Teleop Macros & Manual Control

- **Almost all** of the **teleop** controls are **macros**, which improve **consistency** of each action by **reducing** human error.
- After problems with **localization drift**, we realized that we needed a manual offset for the **turret** and **flywheel**, which ensured other **commands** could still **work reliably**.



Limelight Relocalization

- We noticed that small turret errors and odometry drift in TeleOp caused **huge accuracy issues** for far shots.
- So, we made a command to **correct the robot pose** relative to the **AprilTag on the goal** instead of resetting in the corner.
- This massively **reduced the risk of misfiring**, as we can do it **"on the fly"** after every volley and continue cycling **with confidence**.

Development Process

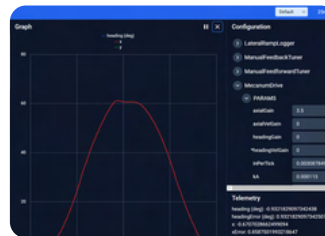
- Since **Scarlet** used the same **coordinate system** as Pedro, we continued using their **visualizer** to **plan pathing** for auto.
- This allowed us to get the **best of both worlds** for Scarlet by **decreasing development time**.

Command Based

- Using **SolversLib**, not only were we able to make **clean and organized** programs with commands, we also could **reuse** autonomous code in teleop, allowing us to **prioritize** auto **performance** and **reliability**.
- So, when auto worked we could be **confident** that teleop would work **well** too.



SolversLib



Programming Efficiency

- To make programming go **faster**, we used several tools created by the FTC community including **FTC Dashboard** and **Sloth Load**
- **FTC Dashboard** allows us to **change** config values on the **fly**, while **Sloth Load** lets us **test** new pieces of code.

```
relocalizer.pose = limelight.getLatest().getBotPose();
resetPose = scarlet.reset(relocalizer.pose);
controllers.bind(VIVEK, LEFT BUMPER, resetPose);
```

